

ASSEMBLY MIDTERM EXAM

1. If $g_2(t) = k g_1(t)$ the energy of the signal $g_2(t)$ equals:

- a) The energy of $g_1(t)$
- b) The energy of $g_1(t)$ multiplied by $2k$
- c) The energy of $g_1(t)$ multiplied by k^2
- d) The energy of $g_1(t)$ multiplied by k

2. The unit impulse response of a linear time invariant system is the unit step function $u(t)$ for $t > 0$, the response of the system to the input $(e^{-at})u(t)$, $a > 0$ will be:

- a) ae^{-at}
- b) $a(1 - e^{-at})$
- c) $(1 - e^{-at})/a$
- d) $1 - e^{-at}$

3. the continuous time system described by:

- a) Causal, linear and time varying.
- b) Non-causal, non-linear and time invariant.
- c) Causal, non-linear and time varying
- d) Non-causal, linear and time invariant.

4. Given a unit step function $u(t)$, its time derivative is:

- a) Unit impulse.
- b) Another step function
- c) A unit ramp function
- d) A sine function

5. An LTI system has impulse response $h(t) = u(t - 1)$, If the input is $x(t) = u(t - 2)$ the output $y(t)$ is given by:

- a) $y(t) = tu(t)$
- b) $y(t) = tu(t - 1)$
- c) $y(t) = (t - 1) u(t - 1)$
- d) $y(t) = (t - 3) u(t - 3)$

6. the fundamental period and frequency of the following signal $x(t) = 2\cos(10\pi t + \pi/6)$:

- a) $T_0=0.1$ sec, $f_0 = 10$ Hz
- b) $T_0 = 0.2$ sec, $f_0 = 5$ Hz
- c) $T_0=10$ sec, $f_0 = 0.1$ Hz
- d) $T_0 = 5$ sec, $f_0 = 0.2$ Hz

7. If $x(t)$ has fundamental period $\tau = 10$ then $x(1/2t - 8)$ has fundamental period:

- a) 5
- b) 20
- c) 10
- d) It is not periodic

8. In these systems any time delay in input signal leads to identical time delay in the output signal:

- a) Time invariant systems
- b) Causal systems
- c) Memory less systems
- d) Non-causal systems

9. Which system is defined as causal system of the following:

- a) $y(n) = 1/3 * x(n) + x(n - 1) + x(n + 1)$
- b) $y(n) = 1/3 * x(n) + x(n - 1) + x[n - (n - 2)]$

11. Which one is discrete time periodic signal:

- a) $\sin \sqrt{3\pi} n$

- b) $\cos \sqrt{2\pi}n$
- c) $\sin 3\pi n$
- d) All of them are correct

12. $Y(t)=x(8t)$ is:

- a) Compressed signal
- b) Expanded signal
- c) Shifted signal
- d) Amplitude scaled signal by a factor of 2

13. Find the content of AL and zero flag (ZF) :after execution of the following code

MOV AL, FEH

ADD AL,2

ADC AL,3

- a) AL = 2 ZF = 1
- b) AL = 6 ZF = 1
- c) AL = 5 ZF=0
- d) AL = 4 ZF = 0 .

14. Defined below X memory locations and the offset address of X is 100h. find the content of register AL after execution of the following code

X DB 88H,44H, 50H, 90H, 55H,23H,51H

MOV BX, OFFSET X

MOV AL, [BX+4]

- a) 55.
- b) 90
- c) 88
- d) 44

15. Which of the following instruction can be used to multiply constants of register DX by 4 :

- a) MOV CL,2; SHR DX,CL
- b) MOV CL,2; RCL DX,CL
- c) MOV CL,2; SHL DX,CL
- d) MOV CL,2; ROL DX,CL

16. One of the following instructions does not affect the destination:

- a) add dest, 1111h
- b) sub dest, 0FFFFh
- c) cmp dest, 1111h
- d) mov dest, 0000h

17. Find the content of AL and zero flag (ZF) after execution of the following code

MOV AL, FEH

ADD AL,2

ADC AL,3

- a) AL = 2 ZF=1
- b) AL = 6 ZF=1
- c) AL = 5 ZF = 0
- d) AL = 4 ZF = 0

18. The unit that controls the external buses and provides access to instructions and data in the 8086 architecture is:

- a) EU
- b) CU
- c) BIU
- d) ALU

19. what is the value of AL register after :executing the following codes

MOV AL, -7

MOV AH,0

MOV BL, 2

DIV BL

- a) 1FH
- b) 7CH.
- c) FDH
- d) 13H

20. The addressing mode used in the : instruction MOV DL,X[SI] is:

- a) Base relative addressing mode
- b) Immediate addressing mode
- c) Direct addressing mode
- d) Direct indexed addressing mode

21. The equivalent code of NOT AX instruction is:

- a) OR AX,AX; ADD AX,1
- b) XOR AX,FFH
- c) XOR AX,FFFFH; ADD AX,1
- d) XOR AX,FFFFH

22. The conditional branch instruction JC.18 performs the operations when if:

- a) CF = 1
- b) CF = 0
- c) PF = 0
- d) ZF = 0

23. Defined below X memory locations and the offset address of X is 100h. Find the content of register AL and DL after execution of the following code

X DB FAH, CBH, 1AH, 80H

MOV BX, OFFSET X

ADD BX,1

MOV AL, [BX]

INC BX

MOV DL, [BX]

a) AL = FAH DL = CBH

b) AL = CBH DL = 1AH

c) AL = FAH DL = 80H

d) AL = 1AH DL = 80H

24. Defined below X memory locations and the offset address of X is 100h. find the content of register AL after execution of the following code

X DB 88H,44H, 50H,90H,55H,23H,51H

MOV BX,OFFSET X

MOV SI,2

MOV AL, [BX][SI]

a) AL=55H

b) AL-50H

c) AL-90H

d) AL-44H

25. what is the value of AL after executing the following code

MOV AL,3

NEG AL

JC L

INC AL

L: DEC AL

- a) FBH
- b) FCH
- c) FDH
- d) 2

26. The binary value of AL and carry flag (CF) after the following instructions have executed is

MOV AL,10101011B

SAR AL,1

- a) AL-11010101B CF=1.
- b) AL-01010101B CF=1
- c) AL-01010100B CF=0
- d) AL=01010100B CF=0

27. what is the value of AL after executing the following

MOV AL,-6

MOV CL,2

SAR AL,CL

INC CL

RCR AL,CL

- a) DFH
- b) A7H
- c) IFH
- d) BFH

28. What is the binary value of AL and CF after? the following instructions have executed

MOV AL,C4H

STC; set CF = 1

RCR AL,1

- a) AL = 62H CF = 0
- b) AL = E2H CF = 0
- c) AL = 62H CF = 1
- d) AL = E2H CF = 1

29. If AX = 1234H and BX=00FFH what is the value stored in AX after the execution of the following instruction

AND AX, BX :

- a) 0034H
- b) 0000H
- c) FF12H

30. Which of the following instructions is the same as IB instruction?

- a) INGE
- b) JNG
- c) JNAE
- d) JNA

31. If the size of memory is 512 bytes, what will be the starting and ending offset addresses of it?

- a) 00H to FFH
- b) 000H to 1FFH
- c) 0000H to FFFFH
- d) 000H to FFFH

32. What is the type of addressing mode used in instruction MOV ALX (SI) :

- a) Indirect
- b) Based relative
- c) Direct Indexed
- d) Direct

33. Address bus width Of Intel 8086 microprocessor is:

- a) 24
- b) 16
- c) 20
- d) 32

34. If the offset address of X in data segment is equal to 215H, and DS-2000H, then the physical address of X =?

- a) 20215 H
- b) 20118H
- c) 2215 H
- d) 30215H

35. Stack pointer (SP) contains offset address of.....segment:

- a) STACK
- b) CODE
- c) DATA
- d) EXTRA segment

36. The content of DL register after execution the instruction MOV D1,-17H :

- a) DL-EAH
- b) BL-FCH
- c) DL- EFH
- d) BL-FEH

37. The equivalent code of NOT AL instruction is:

- a) XOR AL, FFH
- b) AND AL, 00

- c) OR AL, 1
- d) XOR AL, 00H

38. Show the binary value in register AL after execution the following instruction sequence,

MOV AL,11

MOV BL15

SUB AL,BL

- a) AL-11111111
- b) AL-11111011
- c) AL-11111110
- d) AL-11111100

Write an 8085 assembly program to find the total number of the memory byte locations in the memory array named Array in which the bit 2 is set then store it in the byte memory location named count. 14 p Array D6 21H, 84h, OF7h, DE8h, DABH, 8Ch, OC4h, 44h, 74h, 95h count DB 7

implement the following if statement logic using assembler codes:

If $x < 20$ $s = a + 10$ else $s = b + 20$; assuming x,s,a,b signed memory locations.